Personalized Early Sepsis Prediction Using Patient-Specific Physiological Baselines

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***Abstract*—Sepsis is a life-threatening condition frequently encountered in intensive care units (ICUs), where delayed identification substantially increases mortality. Most prior work predicts sepsis directly from raw ICU vital-sign records at the population level. This study instead investigates whether patient-specific physiological baselines improve early sepsis prediction relative to conventional population-level features. Using the PhysioNet/Computing in Cardiology Challenge 2019 dataset, a per-patient baseline was established from the first 12 hours of records, deviations from that baseline were computed over hours 12–24, and sepsis onset between hours 24–48 was used as the prediction target. A total of 10,607 eligible patients were identified, with a 2.97% positive rate. Both a personalized (baseline-deviation) feature set and a conventional (raw population-level) feature set were evaluated using 12 machine learning models under class weighting and SMOTE, with cross-validated threshold tuning. The best conventional model (class-weighted RBF SVM) achieved PR-AUC 0.0775 and ROC-AUC 0.6337, while the best personalized model achieved PR-AUC 0.0768 and ROC-AUC 0.6014. Bootstrap testing found no statistically significant difference between the two representations. We conclude that patient-specific baseline personalization did not outperform conventional population-level features for this prediction task.**

***Index Terms*—sepsis prediction, machine learning, ICU, physiological baseline, class imbalance, PhysioNet**

I. Introduction

Conventional sepsis-prediction systems typically apply fixed clinical thresholds uniformly across all patients. However, physiological states vary substantially between individuals, so a value that is abnormal for one patient may be entirely normal for another. This work develops a personalized early sepsis prediction pipeline that learns each patient's physiological baseline from the first 12 hours of ICU data, measures deviations during the subsequent 12 hours, and predicts sepsis onset during hours 24–48. The objectives are to (i) establish patient-specific baselines, (ii) derive deviation-based features, (iii) train and compare multiple machine learning models, and (iv) evaluate whether personalized features outperform conventional population-level vital-sign summaries using AUROC, PR-AUC, precision, recall, F1, and MCC.

II. Literature Review

Sepsis remains a major cause of death in critically ill patients, motivating substantial work on early prediction. Machine-learning (ML) methods increasingly use routinely collected ICU physiological, laboratory, and clinical data to identify deterioration before clinical onset. Although conventional clinical scores are useful, ML models can capture complex interactions and may provide earlier warning. Most existing approaches, however, learn population-level patterns and do not explicitly represent how a patient's physiology changes from their own normal state.

TABLE I: Summary of representative related studies and their distinction from this work.

Study	Main approach	Limitation relative to this work
Liu et al.	GLM, XGBoost, RNN on ICU data	Population-level pre-shock pattern
Barton et al.	ML using six vital signs	No patient-specific baseline
Rangan et al.	Vital-sign combinations	Uses absolute physiological patterns
Prithula et al.	Classical ML with EHR variables	No explicit baseline deviation
Wang et al.	Multimodal Transformer	Temporal, but not patient-relative
Kim et al.	DeepSEPS with FCN/GRU	Learns population trajectories
Yang et al.	XGBoost and single-point vitals	Single time-point representation
Shao et al.	Nine ML algorithms	Not patient-specific

Liu *et al.* designed an approach to detect a pre-shock state before septic shock using MIMIC-III data and evaluated GLM, XGBoost, and RNN models. Their RNN achieved an AUROC of 0.93 and predicted impending septic shock several hours beforehand [1]. Barton *et al.* evaluated prediction up to 48 hours before sepsis onset using six routinely recorded vital signs and continuously generated a sepsis risk score [2]. However, the patient's individual physiological baseline was not explicitly incorporated.

Rangan *et al.* compared combinations of blood pressure, heart rate, respiratory rate, temperature, and oxygen saturation and found that combinations of vital signs can improve prediction [3]. Prithula *et al.* developed a classical ML framework using routinely collected ICU electronic health-record data, but its features were based on absolute clinical values rather than deviations from an individual's normal state [4].

Wang *et al.* proposed a multimodal Transformer combining physiological time series with clinical notes and reported improvements over LSTM, GRU, and static-embedding baselines

on MIMIC-III and eICU-CRD [5]. The approach models temporal changes but not explicit patient-relative deviations. Shao *et al.* compared nine ML algorithms for early sepsis prediction using MIMIC-III, including XGBoost-DART, Gaussian NB, LightGBM, random forest, and AdaBoost, with temperature reported as an important predictor [8]. Yang *et al.* used single time-point non-invasive vital signs with XGBoost in a multicenter study [7]. Kim *et al.* developed DeepSEPS using FCN feature embeddings and a two-layer GRU for real-time sepsis and septic-shock prediction [6].

A common limitation across these approaches is that the model learns population-level patterns rather than explicitly measuring how abnormal a patient's physiology is relative to that patient's own baseline. This study addresses that gap by using individualized baseline deviation as the predictive representation.

III. Dataset and Preprocessing

A. Dataset

The PhysioNet Challenge 2019 ICU dataset was used, containing 546,123 hourly records across 44 columns for 14,057 patients. Table II summarizes the column groups.

TABLE II: Composition of the ICU record set

Group	Columns	Examples
Vital signs	8	HR, O2Sat, Temp, SBP, MAP, DBP, Resp, EtCO2
Laboratory values	26	BaseExcess, pH, Lactate, WBC, Creatinine
Demographics/admin	6	Age, Gender, Unit1/2, HospAdmTime, ICULOS
Outcome	1	SepsisLabel
Index keys	3	Unnamed: 0, Hour, Patient_ID

The SepsisLabel follows the challenge definition: for septic patients the label is 1 for $t \geq t_{sepsis} - 6$ and 0 otherwise; non-septic patients are labeled 0 throughout. At the record level, class distribution is severely imbalanced (Table III), and among 1,239 septic patients the median detection time is 29 h, motivating a clinically relevant 24–48 h prediction window.

TABLE III: Hourly-record class distribution

SepsisLabel	Records	Percentage
0 (no sepsis)	534,272	97.83%
1 (sepsis)	11,850	2.17%

B. Exploratory Data Analysis

1) Missingness: Missingness dominated design decisions: laboratory columns were largely >90% blank (e.g., EtCO2 100%, Troponin/Fibrinogen >99%), while routine vitals were far more complete (HR 7.71% missing, MAP 10.18%, Resp 9.79%, O2Sat 12.05%, SBP 15.24%, DBP 47.87%, Temp 66.21%). Consequently, only the seven core vitals (HR, SBP, DBP, MAP, Temp, O2Sat, Resp) were retained for baseline construction.

2) Distributions and outliers: Vital-sign distributions were generally unimodal and physiologically plausible, with HR, Resp, and MAP right-skewed and O2Sat left-skewed with a ceiling at 100% (Fig. 1). The 1.5×IQR rule identified

TABLE IV: Selected missingness profile.

Variable	Missing (%)	Variable	Missing (%)
EtCO2	100.00	Temp	66.21
TroponinI	99.2–99.9	DBP	47.87
Lactate	92–97	SBP	15.24
FiO2	86–90	O2Sat	12.05
Glucose	86–90	MAP	10.18
Creatinine	92–97	Resp	9.79
Platelets	92–97	HR	7.71

TABLE V: IQR-based outlier summary for the seven core vital signs.

Feature	Lower	Upper	Outliers (%)
HR	38.50	130.50	4,768 (0.87)
SBP	60.00	180.00	4,212 (0.77)
DBP	28.25	90.25	5,789 (1.06)
MAP	39.32	116.68	7,624 (1.40)
Temp	35.06	39.06	2,758 (0.51)
O2Sat	90.75	104.75	7,976 (1.46)
Resp	4.50	32.50	8,539 (1.56)

4,768 HR outliers (0.87%), 4,212 SBP outliers (0.77%), 5,789 DBP outliers (1.06%), 7,624 MAP outliers (1.40%), 2,758 Temp outliers (0.51%), 7,976 O2Sat outliers (1.46%), and 8,539 Resp outliers (1.56%). These observations were retained because they remain physiologically plausible and may contain deterioration signals. The data were sorted by patient and hour, duplicates were checked, and no duplicate rows were found. StandardScaler was applied only to distance- and gradient-sensitive learners; tree ensembles were left unscaled.

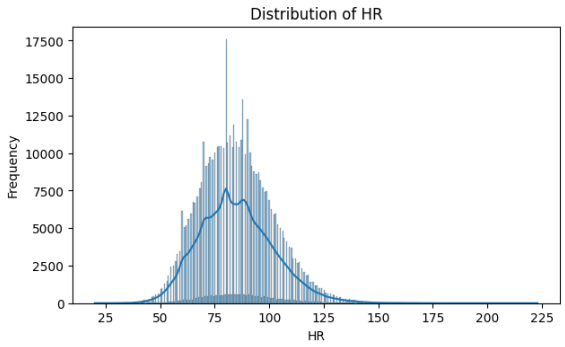


Fig. 1: Distribution of heart rate: unimodal and right-skewed.

3) Group comparison and correlation: Population-level comparisons between septic and non-septic records showed only small mean differences (e.g., HR: 84.82 vs. 90.34 bpm; Resp: 18.71 vs. 20.70 breaths/min), with heavily overlapping distributions (Fig. 2), illustrating the limitation of a single population-wide threshold.

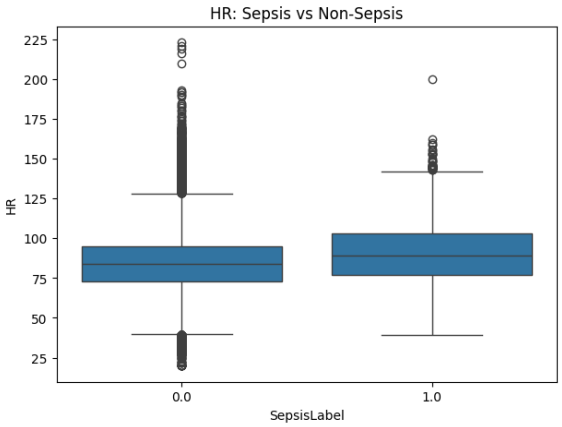


Fig. 2: Heart rate in septic vs. non-septic records: a modest shift within a large overlap.

A Pearson correlation heatmap (Fig. 3) confirmed a haemodynamic block structure, with SBP, DBP, and MAP strongly correlated with one another while HR, Resp, Temp, and O2Sat were relatively independent.

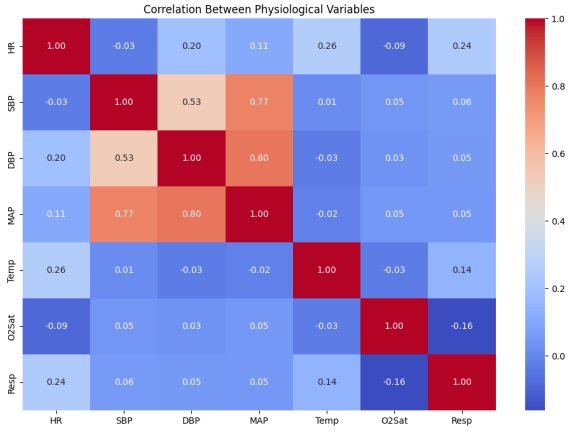


Fig. 3: Correlation heatmap of the seven core vital signs.

4) Temporal exploration: Records were sorted by Patient_ID and Hour, and per-patient trajectories were examined relative to the 12-hour baseline boundary. Fig. 4 shows the distribution of measurement counts per patient alongside a representative heart-rate trajectory, highlighting irregular sampling and non-uniform ICU stay lengths, which necessitated explicit window-eligibility criteria.

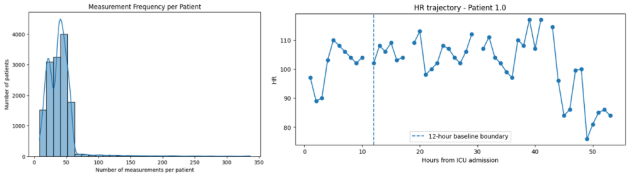


Fig. 4: Measurement frequency per patient (left) and heart-rate trajectory of a single patient with the 12-hour baseline boundary marked (right).

C. Data Preprocessing

Preprocessing proceeded as follows: (1) the seven dense vital signs were retained, discarding EtCO2 and sparse laboratory panels; (2) data were sorted by Patient_ID and Hour for temporally correct windowing; (3) no duplicate rows were found; (4) medians were used for baselines to reduce sensitivity to outliers; (5) no global imputation was applied prior to feature engineering—baselines used NaN-skipping aggregation, and remaining gaps were median-imputed within each scikit-learn pipeline; (6) `StandardScaler` was applied only for distance/gradient-sensitive learners; (7) all hourly features were aggregated into a single row per patient. Leakage was prevented by performing the patient-level 80/20 stratified split *before* any resampling, imputation, scaling, SMOTE, or threshold selection, all of which were confined to the training folds.

1) Leakage Prevention: If the features come from hours 0 to 24 and the label comes from hours 24 to 48, the 80/20 split is stratified at patient level and occurs before, then the pipeline will not leak. `any resampling/tuning` means that no patient is found on both sides, `imputation/scaling/SMOTE` means that imputation and scaling, and SMOTE are executed within the pipeline, and the `decision threshold` is selected by training-fold CV. The test set is never resampled, re-weighted or used to select anything.

D. Handling Class Imbalance

Sepsis positives comprise 2.17% of hourly records and 2.97% of eligible patients (315 positive versus 10,292 negative), approximately 32.7:1 in the training split. Accuracy is therefore not a sufficient objective because a model predicting "no sepsis" for everyone can obtain a superficially high accuracy.

The imbalance was addressed identically in the conventional and personalized arms. First, an 80/20 stratified split was made at patient level before resampling and threshold tuning: 8,485 patients for training (252 positive) and 2,122 for testing (63 positive). Second, cost-sensitive learning used balanced class weights; for boosted models, `scale_pos_weight`=32.67 was used. Third, SMOTE with $k = 5$ was applied only during model fitting. Finally, the decision threshold was selected from five-fold out-of-fold predictions by maximizing F1, and the chosen threshold was applied once to the untouched test set. PR-AUC was used as the primary ranking metric, supported by MCC, balanced accuracy, precision, recall, F1, ROC-AUC, and accuracy.

IV. Methodology

A. Objective

The goal was to use data available through hour 24 of an ICU stay to predict sepsis onset during hours 24–48, and to determine whether a patient-specific baseline representation improves on raw population-level features.

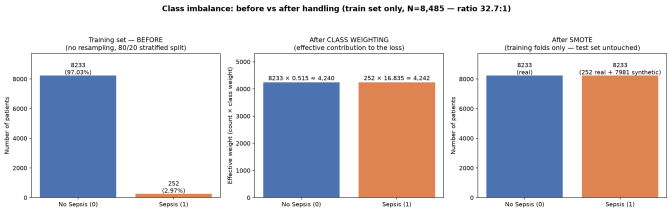


Fig. 5: Class imbalance in the training set before and after cost-sensitive weighting and SMOTE.



Fig. 6: Heart-rate deviation during 12-24 h for septic and non-septic patients in 24-48 h.

B. Three-Window Temporal Design

TABLE VI: Temporal window design

Window	Hours	Purpose
Baseline	$0 \le t < 12$	Establish per-patient vital-sign baseline
Observation	$12 \le t < 24$	Compute personalized deviation features
Outcome	$24 \le t < 48$	Prediction target (sepsis onset)

C. Baseline and Deviation Derivation

For each vital sign v, the patient-specific baseline is

$$\text{baseline}(p, v) = \text{median}\{v(p, t) : 0 \le t < 12\}. \quad (1)$$

Three deviation types were computed for hours 12–24:

$$v_{\text{dev}}(t) = v(t) - \text{baseline}(v) \quad (2)$$

$$v_{\%\text{dev}}(t) = 100 \times \frac{v(t) - \text{baseline}(v)}{\text{baseline}(v)} \quad (3)$$

$$v_{\text{change}}(t) = v(t) - v(t-1) \quad (4)$$

Zero baselines were treated as NaN in the percentage-deviation calculation. Per-patient boxplots of heart-rate deviation showed a higher median deviation among future-septic patients, though overlap remained substantial, indicating a shifting tendency rather than a sharp decision boundary.

D. Patient-Level Feature Formation

Twelve observation hours per patient were summarized into 13 statistics per vital sign (91 total features): baseline level; deviation summary (mean, std, max, min); deviation magnitude (mean/max absolute deviation); relative deviation (mean/max percentage deviation); and rate-of-change (mean, std, max, min).

E. Cohort Definition and Target

A patient was eligible if at least one record existed in each of the three windows. Of 14,057 patients, 10,607 (75.5%) were eligible; 3,450 were excluded for insufficient temporal coverage, independent of class. The target was $\text{Target}(p) = \max\{\text{SepsisLabel}(p, t) : 24 \le t < 48\}$, yielding a $10{,}607 \times 93$ patient-level matrix with 315 positives (2.97%). Only 108 patients had already developed sepsis before hour 24, confirming the task predicts future onset rather than existing sepsis.

F. Conventional Comparison Arm

An analogous conventional feature set (28 features: mean, std, max, min per vital sign over hours 12–24) was built on the identical patients, window, split, imbalance handling, models, and evaluation protocol, differing only in physiology representation. Correlation analysis (Fig. 7) showed the 28-feature conventional matrix forming clear red blocks of near-identical summary statistics, whereas the 91-feature personalized matrix exhibited many smaller blocks corresponding to families of deviation measures per vital sign—reflecting a higher-dimensionality trade-off. The strongest single predictor was Resp_raw_min (conventional) and Resp_baseline (personalized).

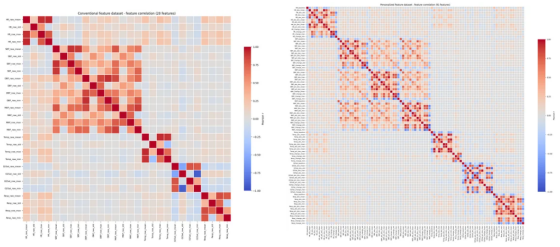


Fig. 7: Correlation structure of the 28 conventional raw-summary features and the 91 personalized baseline-deviation features.

G. Modelling Protocol

We implemented 12 ML models for drawing decision boundary and combined them to verify if the personalized design helps linear, distance based, probabilistic or tree-based learners: Logistic regression: 1) Linear baseline models the log-odds of sepsis as a weighted sum of features for checking linearity. 2) Linear SVM (SGD): fits a maximum margin linear separator by stochastic gradient descent, well-scaled and well-tolerated of the 91-feature matrix. 3) RBF SVM: fits patients into a high dimensional space with a Gaussian kernel, capturing non-linear combinations of deviations while

giving the best output of both arms. 4) Decision tree: well interpretable yet prone to overfitting a number of positive patients. 5)Random forest: many bootstrapped trees on random feature subsets, averaged, shortened the variance of a single tree handling scale-free mixed features. 6) Extra trees: randomly selected splitting points, providing even reduced variance and faster fitting with slightly higher bias. 7) KNN: categorizes a patient by the majority label among its k nearest neighbours, made cost sensitive by distance weighting. 8) Gaussian naive Bayes: probabilistic classifier taking normally distributed and conditionally independent features, very fast and made cost sensitive through tilted class priors. 9) MLP: a tiny neural network that learns non-linear interaction, it achieved the most from personalization. 10) Gradient boosting (XGBoost): builds sequential trees while each correction of each from the previous ones, with `scale_pos_weight` applied for imbalance. 11) Second boosted-tree (LightGBM): another boosting method to check that boosting results are not library dependent. 12) AdaBoost: re-weights misclassified patients while combining many weak learners into a strong learner. Each model was trained under two imbalance strategies (class weighting and SMOTE) on both feature sets, yielding $12 \times 2 \times 2 = 48$ configurations (47 successful). For each configuration, out-of-fold probabilities from stratified 5-fold cross-validation determined the F1-optimal threshold; models were then refit on the full training set and evaluated once on the held-out test set. Metrics: accuracy, balanced accuracy, precision, recall, F1, MCC, ROC-AUC, PR-AUC, and confusion matrix.

H. Champion Selection

The champion model per feature set was selected by highest test PR-AUC, with comparison via metric tables, ROC/PR curves, confusion matrices, bootstrap 95% CIs on ROC-AUC/PR-AUC differences, and an exact McNemar test on thresholded predictions.

V. Results

A. Best-Performing Models

Across 47 fitted models, the class-weighted RBF SVM achieved the highest test PR-AUC on both feature sets (Table VII). Given the 2.97% positivity rate, ROC-AUC correctly ranks the dominant negative class while PR-AUC better reflects ranking of positives; F1 was used for tie-breaking.

TABLE VII: Champion model comparison

Metric	Conventional	Personalized	Diff (P−C)
Threshold (CV-tuned)	0.034	0.035	–
Accuracy	0.7333	0.7549	+0.0217
Balanced accuracy	0.6087	0.5891	−0.0196
Precision	0.0533	0.0511	−0.0022
Recall	0.4762	0.4127	−0.0635
F1	0.0958	0.0909	−0.0049
MCC	0.0835	0.0708	−0.0127
ROC-AUC	0.6337	0.6014	−0.0323
PR-AUC	0.0775	0.0768	−0.0007
TP/FP/FN/TN	30/533/33/1526	26/483/37/1576	–

Both champions produced zero true positives at the default 0.5 threshold; CV-tuned thresholding made them effective, representing the largest practical effect observed in this study (Fig. 8).

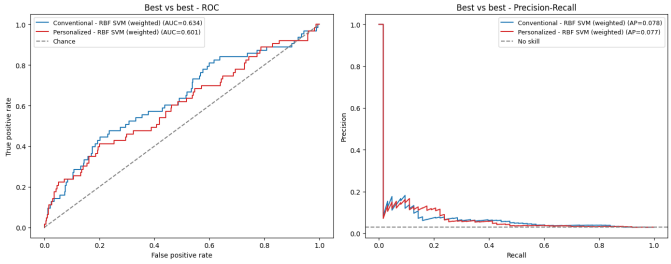


Fig. 8: Best-vs-best ROC and precision–recall curves (RBF SVM, class-weighted).

B. Average Performance by Feature Set and Strategy

Table VIII summarizes mean test performance across all model families. Class weighting outperformed SMOTE on every metric in both arms, and the conventional feature set outperformed the personalized set on average (Fig. 9).

TABLE VIII: Mean test performance by feature set and strategy

Set	Strat.	Acc	Prec	Rec	F1	ROC	PR	MCC
Conv.	Weights	0.813	0.064	0.335	0.102	0.629	0.057	0.079
Conv.	SMOTE	0.869	0.062	0.226	0.085	0.624	0.051	0.060
Pers.	Weights	0.824	0.046	0.250	0.074	0.597	0.047	0.042
Pers.	SMOTE	0.813	0.043	0.208	0.063	0.549	0.042	0.023

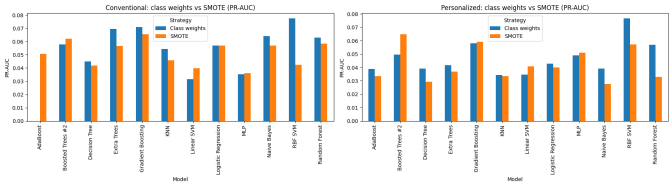


Fig. 9: Class weighting vs. SMOTE (PR-AUC) within both feature sets.

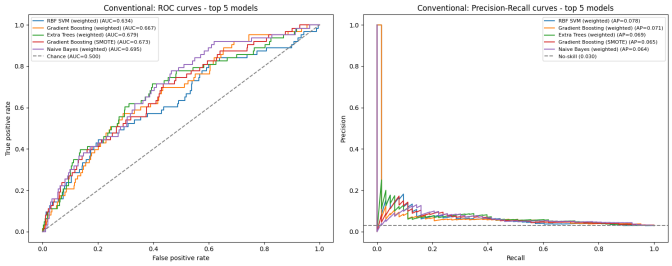


Fig. 10: ROC and precision–recall curves for the top five conventional models.

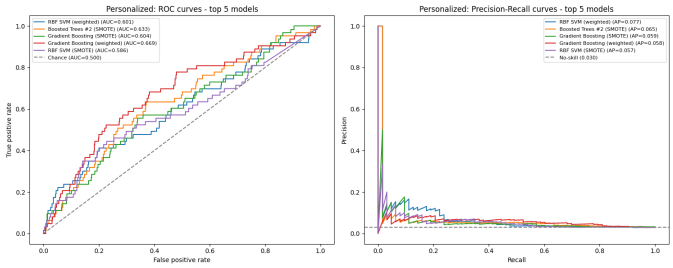


Fig. 11: ROC and precision–recall curves for the top five personalized models.

C. Per-Model Gain from Personalization

Personalization improved ROC-AUC for only 5 of 23 comparable configurations. Flexible, scale-sensitive learners (MLP, linear/RBF SVM) benefited most (e.g., MLP weighted: +0.045 ROC-AUC), while tree ensembles and Naive Bayes degraded substantially (e.g., Random Forest SMOTE: -0.189; Naive Bayes SMOTE: -0.191), suggesting that tree-based models implicitly learn patient-relative thresholds and are hindered by the noisier 91-feature personalized space relative to the more stable 28-feature conventional set.

D. Statistical Significance

Paired bootstrap tests found no significant difference between the two champions: ROC-AUC difference -0.0325 (95% CI [-0.1033, $+0.0342$]); PR-AUC difference -0.0012 (95% CI [-0.0577, $+0.0549$]); one-sided bootstrap $p = 0.512$ against personalized superiority. McNemar's test on thresholded decisions was significant ($p = 0.0197$, 339 vs. 280 discordant pairs), indicating the personalized model achieved marginally higher overall accuracy chiefly through fewer false positives (483 vs. 533) rather than better sepsis detection (26 vs. 30 true positives).

E. Summary

Judged against its hypothesis, the personalized baseline-deviation representation did not outperform the conventional feature design: PR-AUC was statistically tied (0.0768 vs. 0.0775) while the conventional arm led slightly on ROC-AUC and recall. The rigor of the evaluation protocol—patient-level splitting, no test-set resampling, and identical treatment of both arms—supports the validity of this negative result, which may stem from baseline instability under high missingness, deviation noise, low positive density (2.8 positives per feature), and information loss during aggregation.

VI. Conclusion and Future Work

The personalized physiological baseline-deviation representation did not yield improved sepsis prediction in the 24–48 h window relative to conventional raw features (PR-AUC 0.0768 vs. 0.0775; bootstrap CI spanning zero). Two secondary findings emerged: cross-validated threshold tuning is essential, since default 0.5 thresholds yield zero detections under this imbalance, and class weighting consistently outperforms SMOTE at this imbalance ratio. Future work should explore denser and adaptive baseline windows, sequence models (LSTM/GRU/Transformer) that preserve the full deviation trajectory, hybrid conventional-personalized feature combinations, dimensionality reduction, integration of laboratory and demographic context, nested cross-validation, clinically calibrated alert thresholds, and explainability analyses.

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